

PLANNING: Conceitos Fundamentais

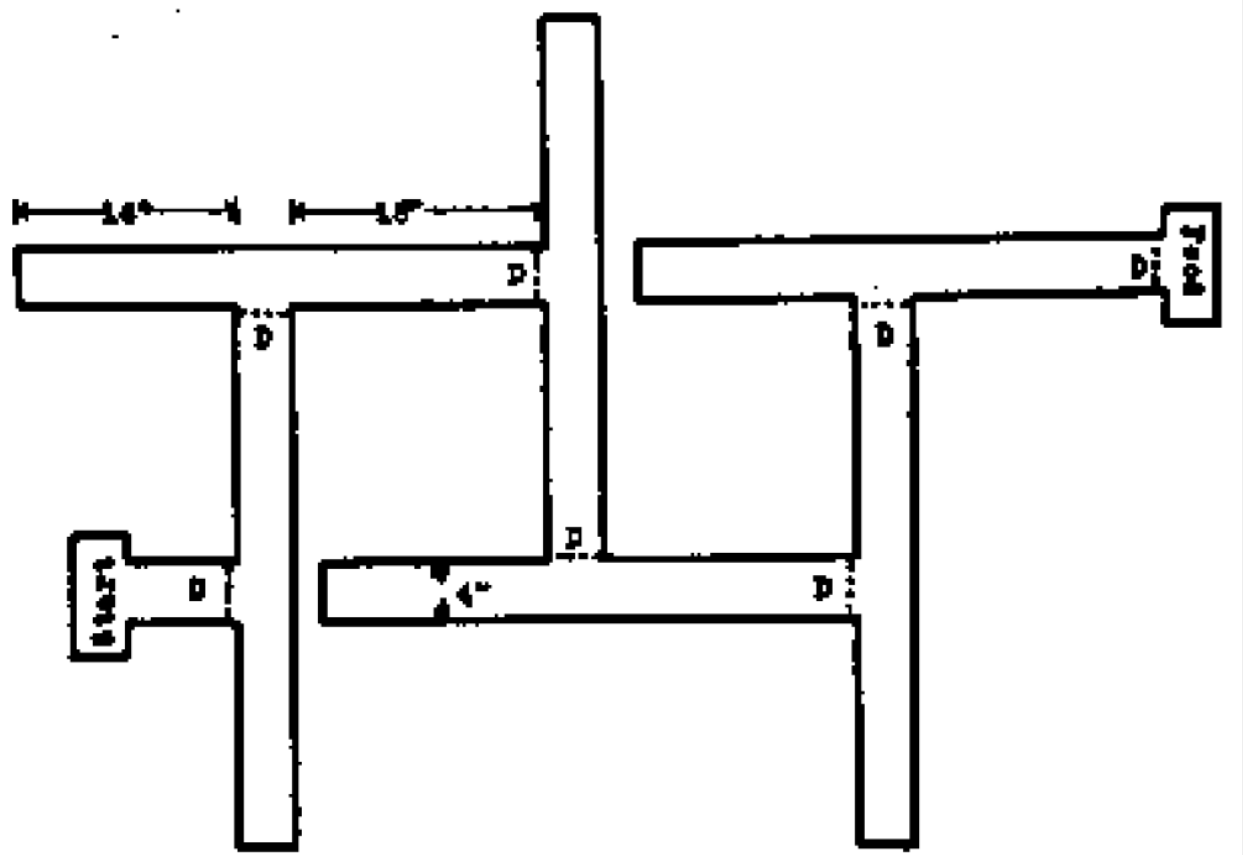
Gustavo Giménez-Lugo

Existe uma Representação do Mundo na “cabeça” ?

É tudo repetição e ação-reação para aprender?

1- Pode existir uma Representação do Mundo na “cabeça” ? [Mapas Cognitivos em Ratos, Tolman, Ritchie & Kalish, 1946]





6-Unit Alley T-Maze

FIG. 4

(From H. C. Blodgett, The effect of the introduction of reward upon the maze performance of rats. *Univ. Calif. Publ. Psychol.*, 1929, 4, No. 8, p. 117.)

SOURCES:

- Classics in the History of Psychology. An internet resource developed by Christopher D. Green. York University, Toronto, Ontario.

->COGNITIVE MAPS IN RATS AND MEN, Edward C. Tolman (1948), First published in The Psychological Review, 55(4), 189-208. http://psychclassics.yorku.ca/Tolman/Maps/maps.htm

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-Rodent image: https://www.freeimages.com/photo/rat-gerbil-hamster-rodent-1356511

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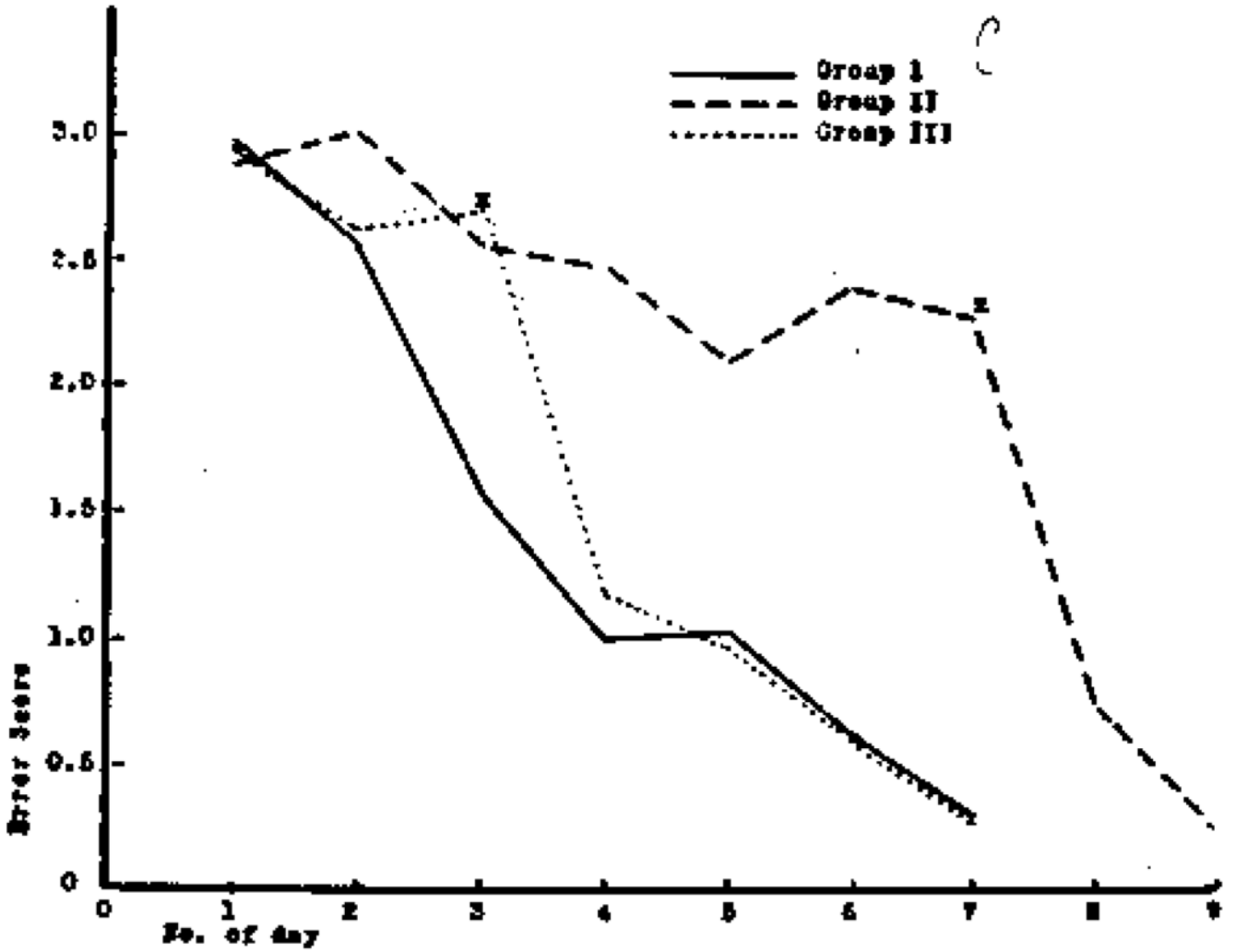


FIG. 5

(From H. C. Blodgett, The effect of the introduction of reward upon the maze performance of rats. *Univ. Calif. Publ. Psychol.*, 1929, 4, No. 8, p. 120.)

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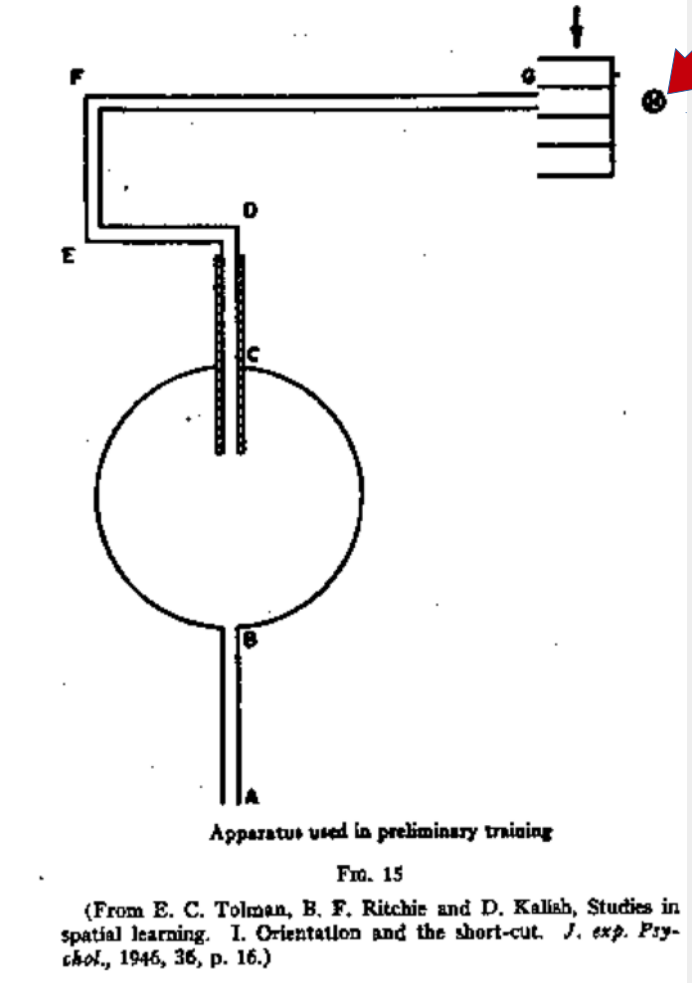
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Apparatus used in preliminary training

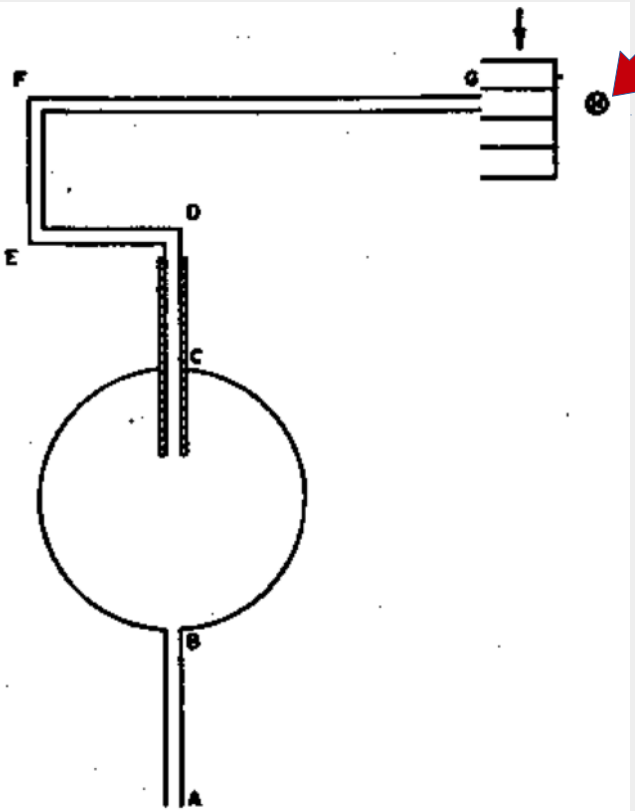
FIG. 15

(From E. C. Tolman, B. F. Ritchie and D. Kalish, Studies in spatial learning. I. Orientation and the short-cut. *J. exp. Psychol.*, 1946, 36, p. 16.)

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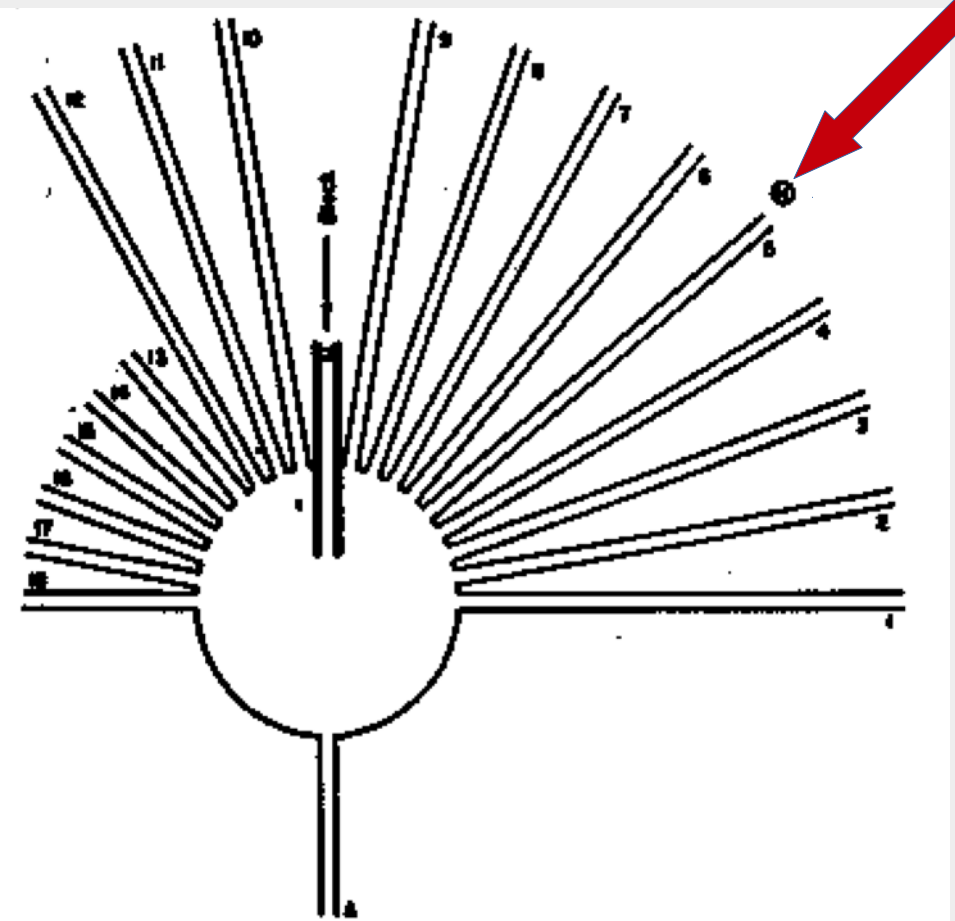
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Apparatus used in the test trial

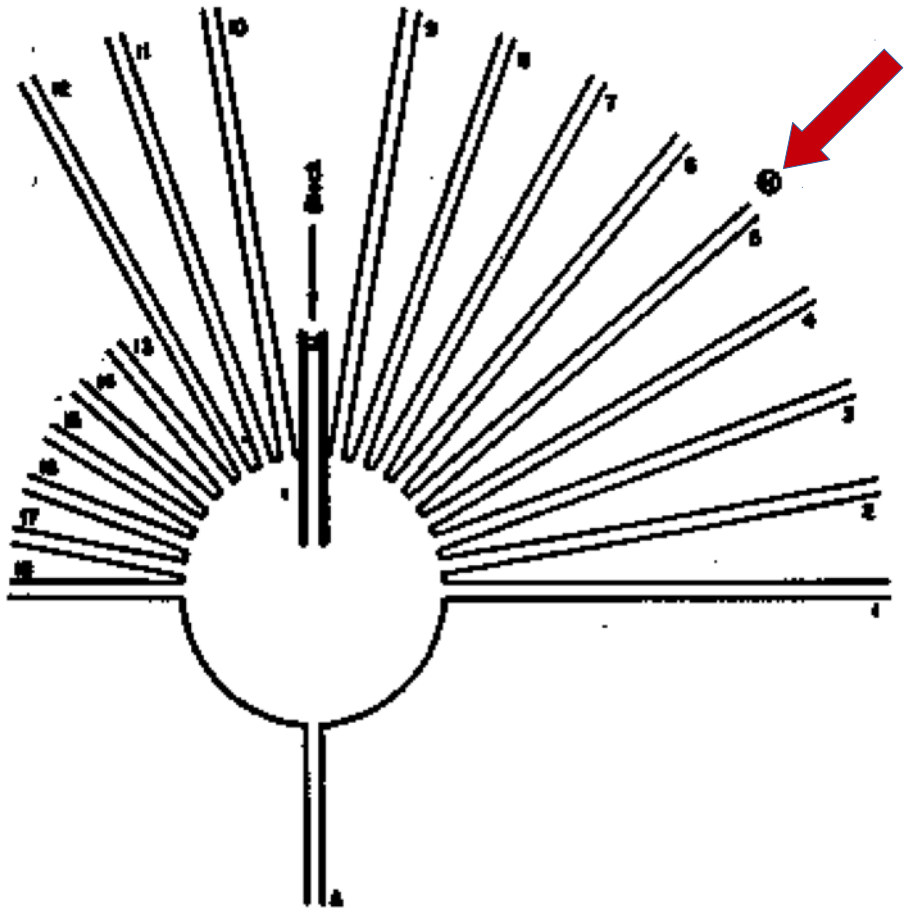
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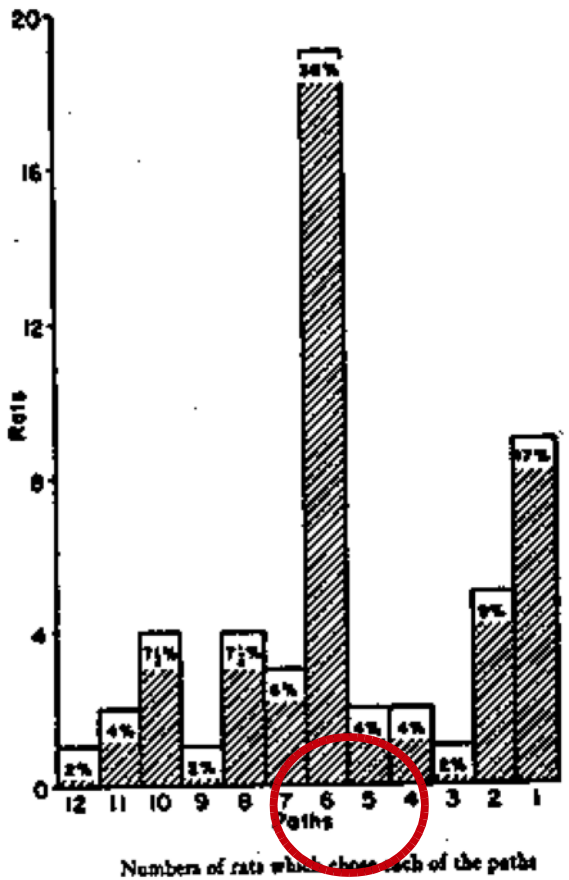


FIG. 17

(From E. C. Tolman, B. F. Ritchie and D. Kalish, Studies in spatial learning. I. Orientation and the short-cut. *J. exp. Psychol.*, 1946, 36, p. 19.)

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2- Pode-se usar a Representação para IMAGINAR/PROJETAR/"PREVER" o que pode acontecer no mundo OBSERVADO?

Funt, B.V., 'Problem-Solving with Diagrammatic Representations', Artificial Intelligence, Vol. 13 No. 4 1980, pp. 201-230.
https://www.cs.sfu.ca/~funt/ProblemSolvingWithDiagrammaticRepresentations_AIJournal1980.pdf

2- Pode-se usar a Representação para IMAGINAR/PROJETAR/"PREVER" o que pode acontecer no mundo OBSERVADO? (WHISPER, a high level reasoner, computer problem solving testing stability of a “BLOCKS WORLD”)

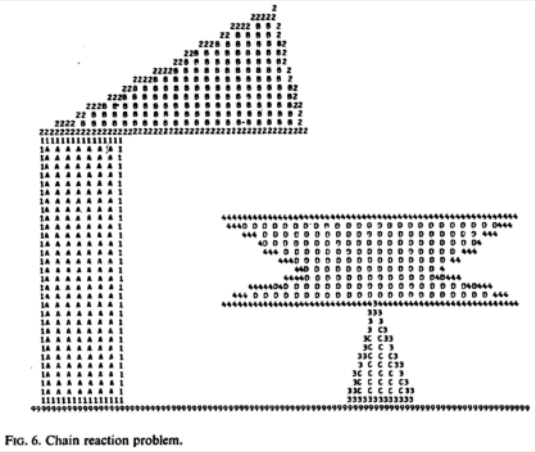


FIG. 6. Chain reaction problem.

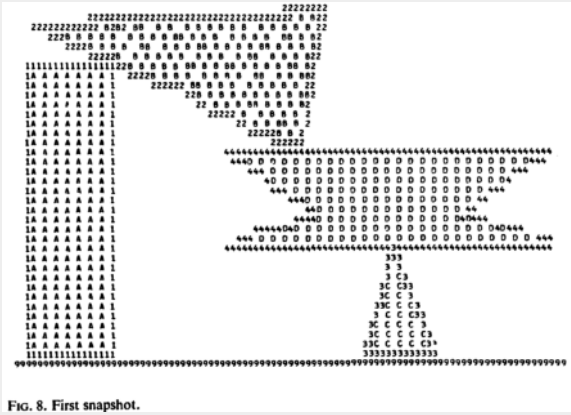


FIG. 8. First snapshot.

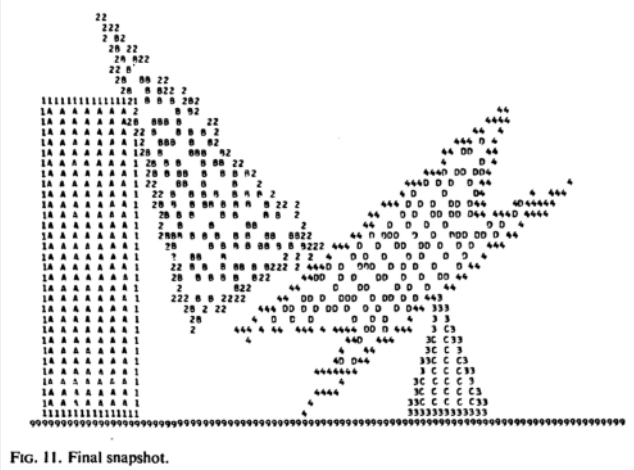
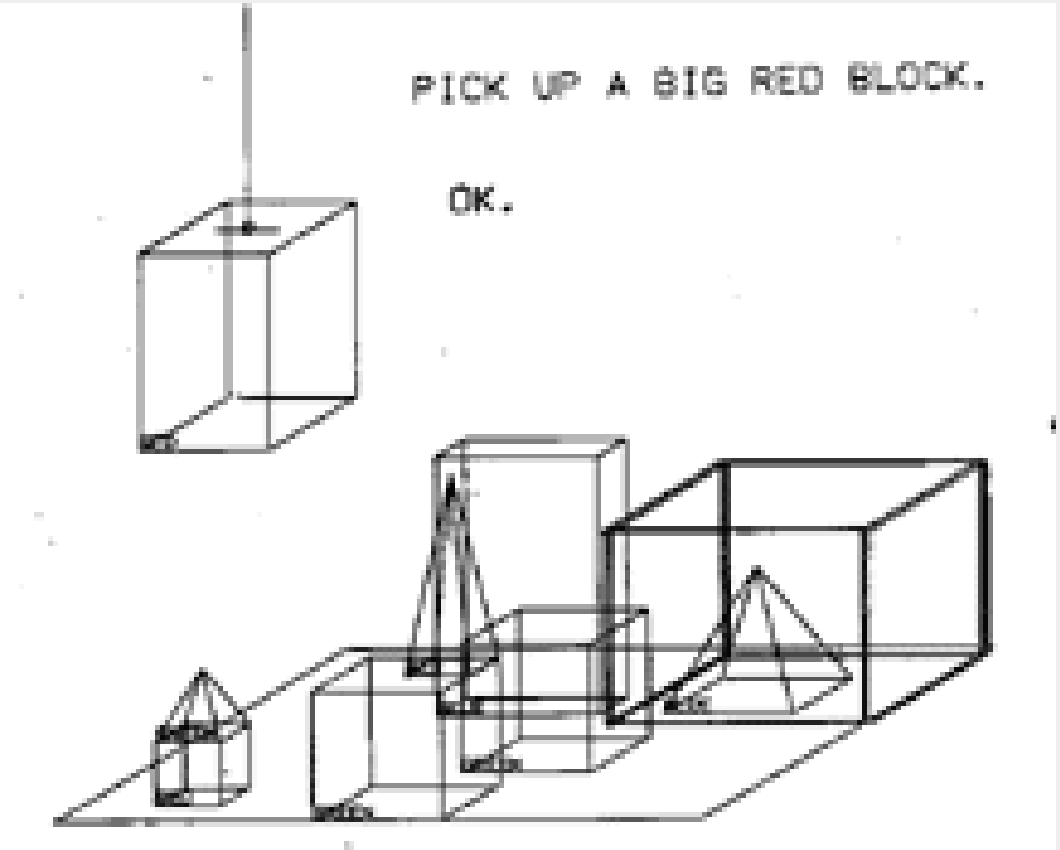


FIG. 11. Final snapshot.

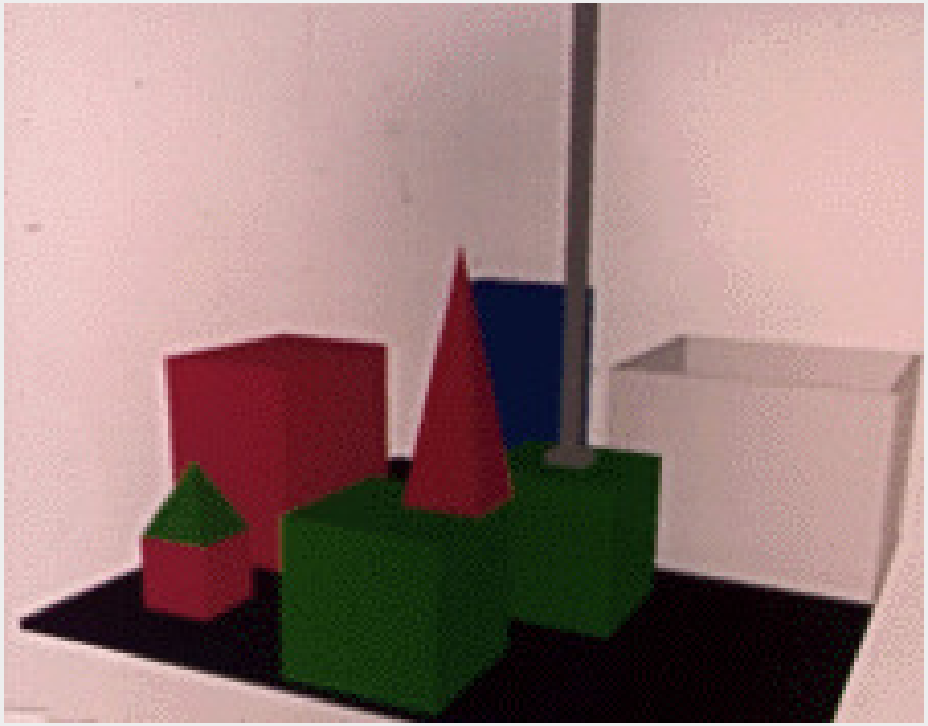
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https://www.cs.sfu.ca/~funt/ProblemSolvingWithDiagrammaticRepresentations_AIJournal1980.pdf

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ?

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU)



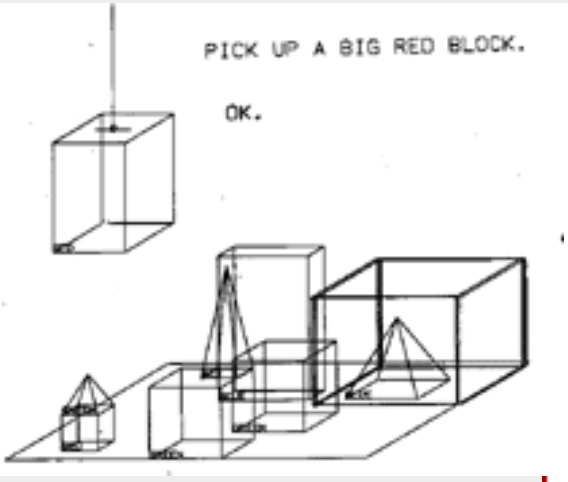
Original screen display

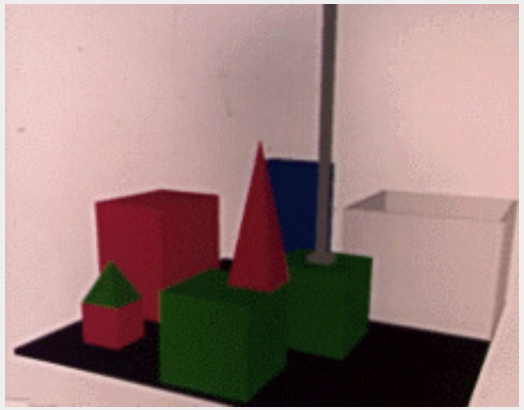


Later color rendering (Univ. of Utah)

SOURCE:
Terry Winograd's dissertation, which was issued as MIT AI Technical Report 235, February 1971 with the title Procedures as a Representation for Data in a Computer Program for Understanding Natural Language. Published as a full issue of the journal Cognitive Psychology Vol. 3 No 1, 1972, and as a book, Understanding Natural Language (Academic Press, 1972).
http://hci.stanford.edu/winograd/shrdlu/

3- Pode-se IMAGINAR o que pode acontecer se EU FAÇO ALGO no MUNDO ? (SHRDLU, 1968-1970 a program for understanding natural language)





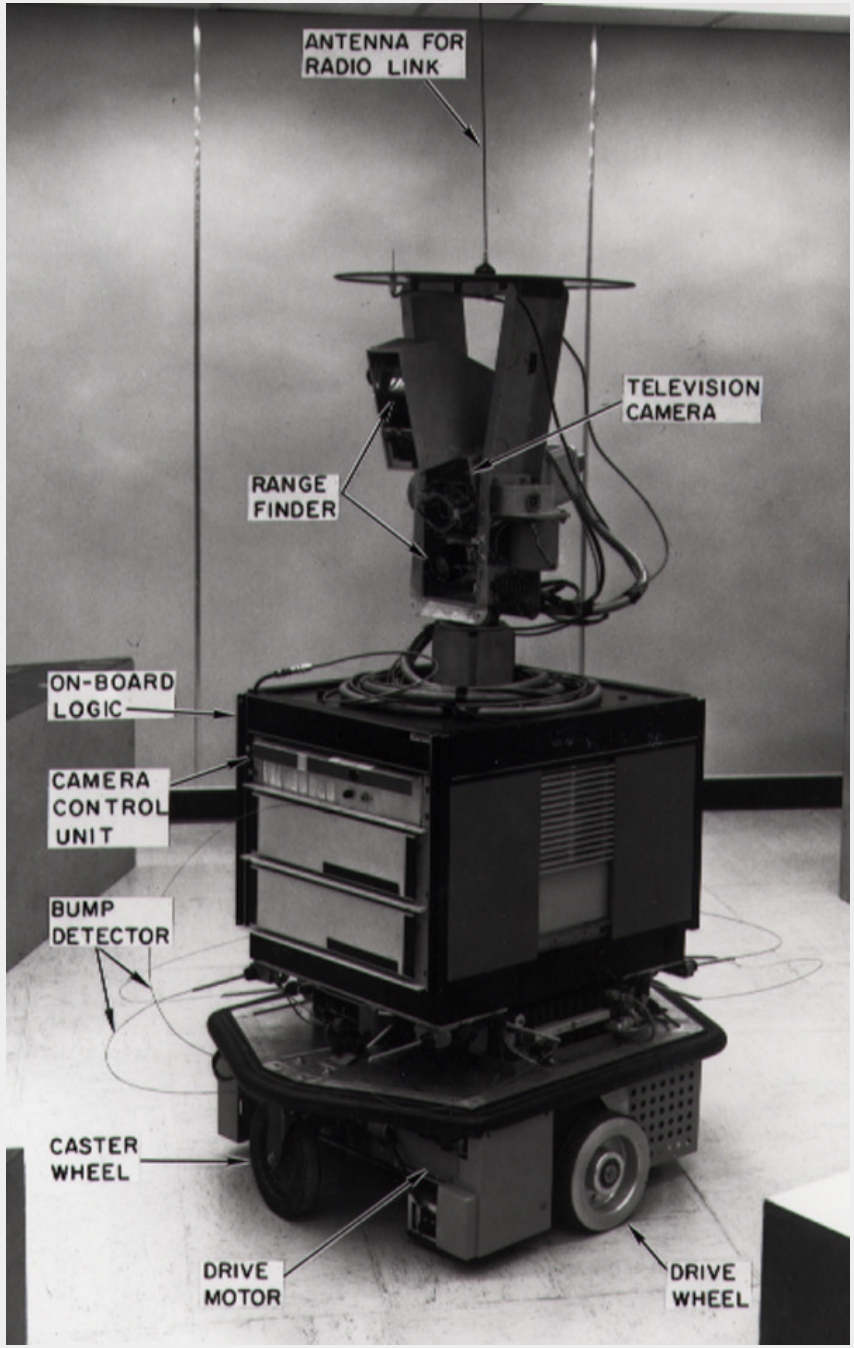
The dialog that was used as a SHRDLU demo:

```
Person: PICK UP A BIG RED BLOCK.
Computer: OK. (does it)
Person: GRASP THE PYRAMID.
Computer: I DON'T UNDERSTAND WHICH PYRAMID YOU MEAN.
Person: FIND A BLOCK WHICH IS TALLER THAN THE ONE YOU ARE HOLDING AND PUT IT INTO THE BOX
Computer: BY "IT", I ASSUME YOU MEAN THE BLOCK WHICH IS TALLER THAN THE ONE I AM HOLDING.
Computer: OK. (does it)
Person: WHAT DOES THE BOX CONTAIN?
Computer: THE BLUE PYRAMID AND THE BLUE BLOCK.
Person: WHAT IS THE PYRAMID SUPPORTED BY?
Computer: THE BOX.
Person: HOW MANY BLOCKS ARE NOT IN THE BOX?
Computer: FOUR OF THEM.
Person: IS AT LEAST ONE OF THEM NARROWER THAN THE ONE WHICH I TOLD YOU TO PICK UP?
Computer: YES, THE RED CUBE.
Person: IS IT SUPPORTED?
Computer: YES, BY THE TABLE.
Person: CAN THE TABLE PICK UP BLOCKS?
Computer: NO.
Person: CAN A PYRAMID BE SUPPORTED BY A BLOCK?
```

SOURCE:
SHRDLU is described in Terry Winograd's dissertation, which was issued as MIT AI Technical Report 235, February 1971 with the title Procedures as a Representation for Data in a Computer Program for Understanding Natural Language It was published as a full issue of the journal Cognitive Psychology Vol. 3 No 1, 1972, and as a book, Understanding Natural Language (Academic Press, 1972).
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4- Pode-se IMAGINAR e depois FAZER algo no MUNDO … e depois ainda atualizar crenças sobre o MUNDO ?

4- Pode-se IMAGINAR e depois FAZER algo no MUNDO … e depois ainda atualizar crenças sobre o MUNDO ? (SHAKEY, 1966-1972)



"SHAKEY was an important development because the RESEARCH concentrated on WHAT the ROBOT THOUGHT and NOT on WHAT it DID" (01m00s)

"SHAKEY foi um desenvolvimento importante porque a PESQUISA estava interessada NO QUE O ROBÔ PENSAVA, NÃO no que FAZIA" (01m00s)

Shakey The Robot
https://vimeo.com/5072714

[Shakey was developed at the Artificial Intelligence laboratory of the Stanford Research Institute in 1966 through 1972 with Charles Rosen as project manager. Major contributors included Nils Nilsson, Alfred Brain, Bertram Raphael, Richard Duda, Peter Hart, Richard Fikes, Richard Waldinger, Thomas Garvey, Jay Tenenbaum, and Michael Wilber. Programming was primarily in LISP.

Story by Scott McGrew. Directed by Barton Bishoff. Additional photography by Donald Hardy and John Chiala.]

Shakey em 1972
SOURCE:
https://en.wikipedia.org/wiki/Shakey_the_robot
#/media/File:SRI_Shakey_with_callouts.jpg
https://vimeo.com/5072714
https://www.youtube.com/watch?v=7bsEN8mwUB8